# MANEESH ANAND


Data Engineer | Data Analyst | Python Developer | Machine Learning Engineer
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## Professional Summary

Data professional with a Master's in Computer Science and hands-on experience in data engineering, analytics, machine learning, and backend development. Skilled in Python, SQL, Power BI, AWS, ETL pipelines, data modeling, dashboard development, machine learning, and REST API development. Experienced in building scalable data workflows, SQL reporting solutions, AI-enabled applications, and production-ready Python systems.

## Technical Skills

**Languages:** Python, SQL, R, JavaScript
**Data Engineering:** ETL Pipelines, Data Modeling, Data Warehousing, Data Cleaning, Data Transformation, Data Quality, Data Integration
**Analytics & BI:** Power BI, Tableau, Excel, Power Query, DAX, KPI Reporting, Dashboard Development
**Databases:** MySQL, PostgreSQL, MongoDB, SQL Server
**Machine Learning:** TensorFlow, Scikit-learn, OpenCV, CNNs, Regression, Classification, Feature Engineering, Model Evaluation
**Backend Development:** Django, Flask, FastAPI, REST APIs
**Cloud & Tools:** AWS S3, EC2, Lambda, Redshift, Glue, Git, Docker, Linux, Jupyter Notebook
**Modern Data Stack:** Databricks, PySpark, Snowflake, dbt, Apache Airflow

## Professional Experience

**Project Team Lead** Oct 2024 – Feb 2025
*University of Central Missouri, Missouri, United States*
- Led development of an AI-powered parking management platform using Python, Django, Flask, MySQL, HTML, CSS, and JavaScript.
- Designed normalized database schemas for vehicle tracking, slot allocation, billing, reporting, and user management workflows.
- Built RESTful APIs for real-time parking availability, booking, payment processing, and admin dashboard operations.
- Developed analytics dashboards to monitor occupancy, revenue trends, usage patterns, and peak-hour traffic.
- Automated manual tracking and reporting workflows, improving operational efficiency and reducing repetitive administrative tasks.
- Managed testing, debugging, deployment preparation, database optimization, and application performance improvements.

**Machine Learning Engineer Intern** Jul 2023 – Nov 2023
*Cognibot, Chennai, India*
- Designed computer vision pipelines using OpenCV and TensorFlow for image classification and pattern recognition tasks.
- Built regression and classification models to support predictive analytics, KPI forecasting, and business decision-making.
- Performed data preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation using Python.
- Evaluated model performance using accuracy, precision, recall, F1-score, RMSE, and validation-based comparison methods.
- Automated machine learning workflows using Scikit-learn, Jupyter Notebook, reusable scripts, and structured datasets.
- Collaborated with technical teams to improve model performance, code quality, and deployment readiness.

**Data Analyst** Oct 2022 – May 2023
*Brane Enterprises, Hyderabad, India*
- Analyzed structured and unstructured datasets to support reporting, business analysis, and data-driven decision-making.
- Developed SQL queries for data extraction, transformation, validation, cleansing, aggregation, and recurring reporting.
- Built KPI dashboards and performance reports to track operational, financial, and business metrics.
- Cleaned, transformed, and validated raw datasets to improve reporting accuracy, consistency, and usability.
- Improved reporting workflows through query optimization, reusable data models, and automated reporting processes.
- Supported stakeholders with trend analysis, ad hoc reports, business insights, and performance monitoring.

## Projects

**EZ Parking Lot Management System with AI Chatbot**
- Built a full-stack parking management application using Django, Flask, MySQL, HTML, CSS, and JavaScript.
- Integrated an AI chatbot to support real-time booking assistance, slot availability, and customer interaction.
- Designed REST APIs for parking allocation, payment workflows, user management, and admin operations.
- Created analytics modules to monitor occupancy rates, revenue trends, booking activity, and system usage.
- Implemented secure authentication and role-based access control for admins, staff, and users.

**Real-Time Sign Language Recognition**
- Developed a CNN-based real-time gesture recognition system using TensorFlow, OpenCV, and Python.
- Built live camera feed processing pipeline for low-latency image classification and gesture prediction.
- Applied image preprocessing, resizing, normalization, and augmentation techniques to improve model accuracy.
- Trained and evaluated deep learning models using classification metrics and validation datasets.
- Optimized inference workflow to support real-time recognition and smoother user interaction.

**Forest Cover Type Prediction**

- Built supervised machine learning models using Python and Scikit-learn to classify forest cover types.
- Performed exploratory data analysis, feature engineering, preprocessing, and dataset validation.
- Applied Logistic Regression, Random Forest, XGBoost, and classification-based modeling techniques.
- Used cross-validation and evaluation metrics to improve prediction accuracy and model reliability.
- Created visual analysis outputs to communicate model performance, feature impact, and classification results.

## Education

**Master of Science in Computer Science** University of Central Missouri, United States

## Certifications

Google Advanced Data Analytics Professional Certificate | AWS Machine Learning Certification | Microsoft Power BI PL-300 – In Progress